

```
import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression, LinearRegression
from sklearn.metrics import r2_score
```

```
import numpy as np
import pandas as pd

X = np.array([1.1,1.3,1.5,2.0,2.2,2.9,3.0,3.2,3.7,4.0,2.9])

Y = np.array([39000,46000,37700,39800,56600,60100,np.nan,54400,57100,63200,56600])

df = pd.DataFrame({
    "Experience":X,
    "Salary":Y
})

print(df)
```

```
   Experience  Salary
0         1.1  39000.0
1         1.3  46000.0
2         1.5  37700.0
3         2.0  39800.0
4         2.2  56600.0
5         2.9  60100.0
6         3.0     NaN
7         3.2  54400.0
8         3.7  57100.0
9         4.0  63200.0
10        2.9  56600.0
```

```
plt.plot(X,Y, marker="D")

plt.xlabel("Experience")
plt.ylabel("Salary")
plt.title("original Data - Experience vs Salary")
plt.grid()
plt.show()
```



```
df.isnull()
```

|    | Experience | Salary |
|----|------------|--------|
| 0  | False      | False  |
| 1  | False      | False  |
| 2  | False      | False  |
| 3  | False      | False  |
| 4  | False      | False  |
| 5  | False      | False  |
| 6  | False      | True   |
| 7  | False      | False  |
| 8  | False      | False  |
| 9  | False      | False  |
| 10 | False      | False  |

```
X = df[["Experience"]]
Y = df["Salary"]
```

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=1)
```

```
X_test.shape
```

```
(3, 1)
```

```
Y_train.shape
```

```
(8,)
```

```
Y_test.shape
```

```
(3,)
```

```
X_train.shape
```

```
(8, 1)
```

```
model = LinearRegression()
```

```
Y_train_cleaned = Y_train.fillna(Y_train.mean())
model.fit(X_train, Y_train_cleaned)
```

```
LinearRegression
```

```
LinearRegression()
```

```
Y_pred = model.predict(X_test)
print("prediction value", Y_pred)
```

```
prediction value [45091.02530961 48528.80877964 49903.92216765]
```

```
r2 = r2_score(Y_test, Y_pred)
print("R2 score", r2)
```

```
R2 score 0.18154498100942407
```

```
experience = np.array([[5]])
```

```
salary = model.predict(experience)
print("Predicted salary for 5 years Experience:", salary[0])
```

```
Predicted salary for 5 years Experience: 69155.50959979321
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does not have valid feature names, but
warnings.warn(
```

```
plt.plot(X, Y, label="Actual Salary")
plt.plot(X, model.predict(X), label="Regression Line")

plt.xlabel("Years of Experience")
plt.ylabel("Salary")
plt.title("Liner Regression - Salary vs Experience")
plt.legend()
plt.grid()
plt.show()
```



```
df.duplicated()
```

```
0
0 False
1 False
2 False
3 False
4 False
5 False
6 False
7 False
8 False
9 False
10 False
```

```
dtype: bool
```

```
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
Experience  0
Salary      1
dtype: int64
```

```
print("\nDuplicate Rows:")
print(df.duplicated().sum())
```

```
Duplicate Rows:
0
```

```
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())  
df
```

|    | Experience | Salary  |
|----|------------|---------|
| 0  | 1.1        | 39000.0 |
| 1  | 1.3        | 46000.0 |
| 2  | 1.5        | 37700.0 |
| 3  | 2.0        | 39800.0 |
| 4  | 2.2        | 56600.0 |
| 5  | 2.9        | 60100.0 |
| 6  | 3.0        | 51050.0 |
| 7  | 3.2        | 54400.0 |
| 8  | 3.7        | 57100.0 |
| 9  | 4.0        | 63200.0 |
| 10 | 2.9        | 56600.0 |